

have detected an effect that converts this heat into a quantum mechanical phenomenon—known as spin—in a semiconductor. Once developed, the effect could enable integrated circuits that run on heat.

In the experiment, the researchers heated one side of the sample, and then measured the orientations of spins on the hot side and the cool side. On the hot side, the electrons were oriented in the spin-up direction, and on the cool side, they were spin-down.

Phone that charges wirelessly

Japan's NTT DoCoMo has demonstrated the prototype of a cellphone that is based on the emerging wireless charging standard. The new standard is likely to replace the conventional power-brick adaptors used for charging portable gadgets. According to NTT DoCoMo, charging time is about the same as a wired charger's.

The Qi charger transfers power by induction. Electricity flows when a wire coil in the cellphone is aligned with an energised coil in a charging pad. The charging pad has a single coil that moves to align itself with the cellphone that is placed upon it. When two phones are placed upon it, the charging coil aligns itself with the first phone and, when charging is complete, it moves over to charge the second phone.



NTT DoCoMo's Qi wireless charger

In the DoCoMo demonstration, the charging pad was built into a table top and the cellphone was placed on the pad for charging to begin.

In addition to the cellphone with an embedded coil, DoCoMo showed the prototypes of a cellphone battery with an integral coil, which could be fitted into cellphones that don't have the required coil to allow wireless charging, and a portable back-up battery pack with the coil.

Smartphones to get eyes

Qualcomm, BlackBerry Partners Fund, US electronics retail chain Best Buy and Anthem Venture Partners are putting roughly \$10 million into Viewdle—a technology startup specialising in the field of augmented reality. Viewdle is reportedly crafting ways to let smartphones see things the same way as people do and identify faces. It has already released a test version of the software which lets desktops recognise people in pictures while uploading on Facebook. Viewdle's technology allows smartphones to recognise people as soon as cameras take pictures or videos showing their faces.

"The first application is face recognition, but the real opportunity is the ability of the machine to take action depending on what it sees in the environment around it. This is similar to how the human brain works," says

Laurent Gil, co-founder and CEO, Viewdle.

Machine learning is the enabling technology, with users teaching computers what their friends look like. That recognition is then shared between devices over the Internet. So if you have a picture of a friend on your desktop computer, it will tell your mobile phone. By the time you take a picture with the phone, it knows it is your friend when the image is captured.



Pandigital's Novel e-reader

Multifunction e-reader with full touchscreen

Pandigital has come up with a multifunction e-reader that includes a touchscreen for easy navigation, a 15cm (6-inch) electrophoretic e-paper display for enjoyment in bright sunlight and a battery life that lasts thousands of page-turns. Made by Sipex/AUO for Pandigital, the display ensures that users get a first-rate reading experience. In addition to being fun and easy to use, it is non-reflective, so the device is easy to read in bright light and sunlight.

The e-reader, called the Pandigital Novel, also allows you to browse the Web, check e-mails, view photos or listen to music. The built-in Wi-Fi and Web browser let you go online. You can enjoy your favourite music via the built-in speaker or use the stereo-mini headphone jack for private enjoyment.

The 2GB internal memory allows you to store hundreds of books and periodicals, music, photos and more. Plus, the built-in memory card reader can accept up to 32GB memory cards for access to a very large number of files.

Laptops may toast your skin

If you use laptop on your lap for long hours, it can discolour your thighs and